Design and Implementation of Signed 8-bit Radix-4 Booth Recoded Multiplier

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Abstract

This report presents the design and implementation of a signed 8-bit Radix-4 Booth Recoded Array Multiplier using behavioral Verilog. The design targets the Xilinx Artix-7 FPGA (XC7A35T-CPG236C, Speed Grade -1). Partial products are generated concurrently and reduced using a Carry-Save Adder (CSA) tree followed by a final carry-propagate addition. Functional simulation, synthesis, implementation, and timing analysis were performed to evaluate area utilization and delay performance.

I. Introduction

Multiplication is a fundamental arithmetic operation in digital systems. Radix-4 Booth recoding reduces the number of partial products by grouping multiplier bits in overlapping triplets. This improves performance compared to conventional array multipliers.

The objective of this lab is to design, simulate, synthesize, and analyze the performance of an 8-bit signed Radix-4 Booth multiplier on Artix-7 FPGA.

II. Theory of Radix-4 Booth Multiplication

Radix-4 Booth recoding examines three bits of the multiplier at a time and generates partial products according to the following encoding:

Booth Code	Operation
000, 111	0
001, 010	+A
011	+2A
100	-2A
101, 110	-A

TABLE I: Radix-4 Booth Recoding Table

For an 8-bit multiplier, only four partial products are generated.

III. Architecture and Block Design

The architecture consists of:

- Input Registers
- Booth Recoder
- Partial Product Generator
- Carry-Save Adder Tree
- Final Carry-Propagate Adder
- Output Register

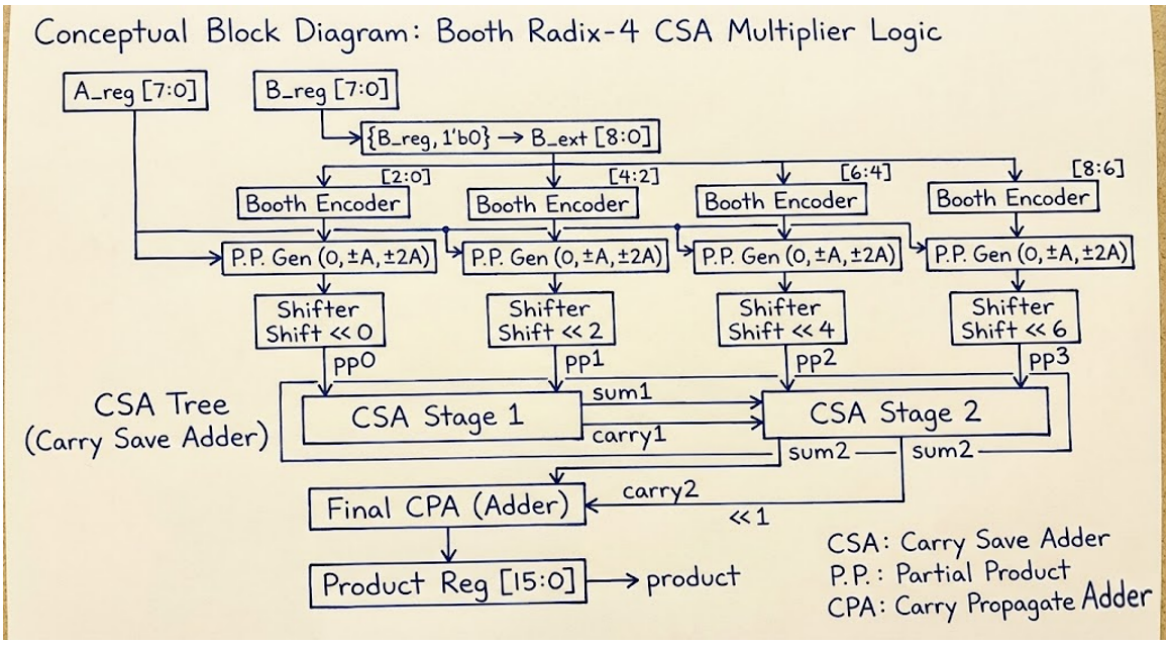


Fig. 1: Block Diagram of 8-bit Radix-4 Booth Multiplier

IV. Verilog Implementation

The multiplier was coded behaviorally in Verilog with:

- Booth recoding function
- Sign-extended partial products
- CSA-based partial product reduction
- Final carry-propagate adder

Input and output registers were included as specified.

V. Simulation

The design was verified using a comprehensive testbench including corner cases, exhaustive testing (-8 to 7), and 3000 random test cases.



Fig. 2: Behavioural Simulation

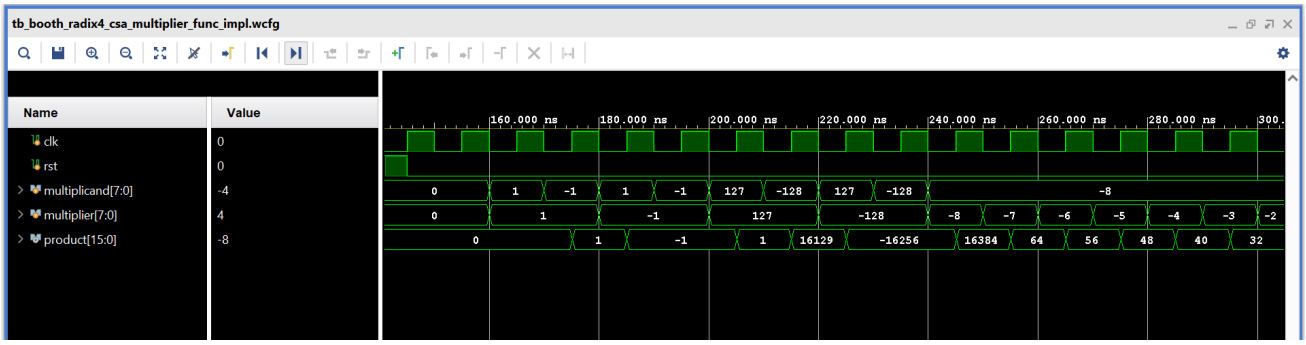


Fig. 3: Functional Simulation Waveform (Post-synthesis)

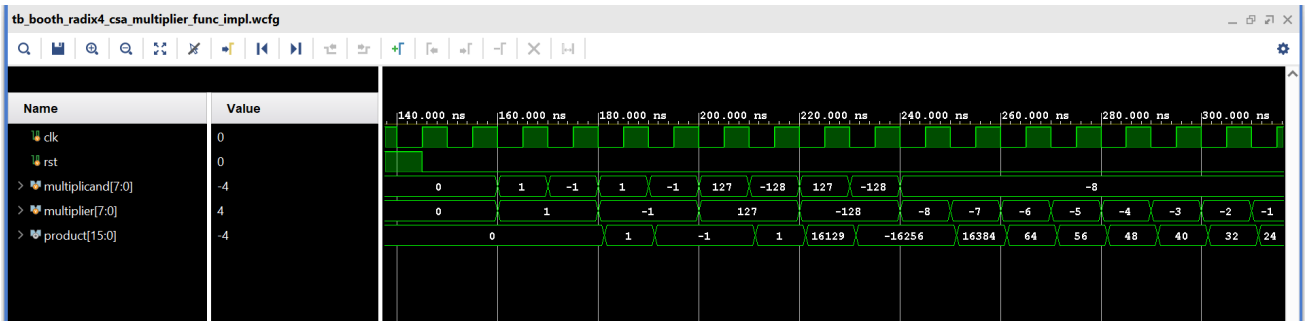


Fig. 4: Timing Simulation Waveform (Post-synthesis)



Fig. 5: Functional Simulation Waveform (Post-implementation)

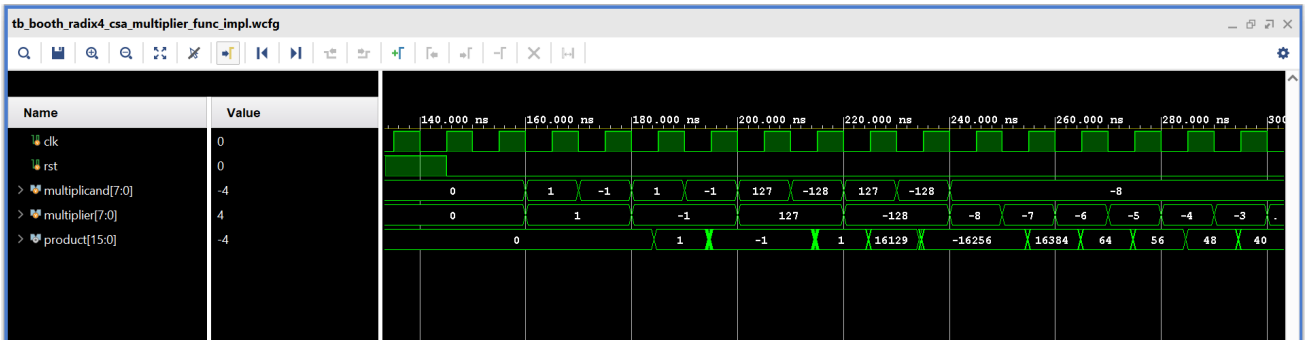


Fig. 6: Timing Simulation Waveform (Post-implementation)

VI. RTL Schematic

After elaboration, the RTL schematic confirms correct structural mapping of Booth recoding logic and CSA tree.

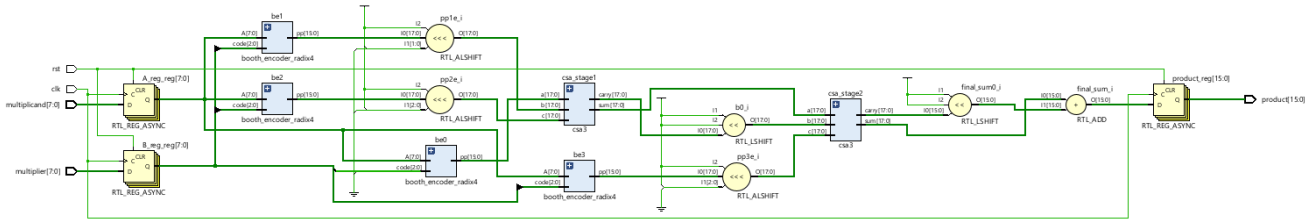


Fig. 7: RTL Schematic

VII. Resource Utilization Results

The design was synthesized targeting Xilinx Artix-7 XC7A35T-CPG236C (-1).

Utilization Post-Synthesis | **Post-Implementation**

Graph | **Table**

Resource	Utilization	Available	Utilization %
LUT	75	20800	0.36
FF	32	41600	0.08
IO	34	106	32.08
BUFG	1	32	3.13

Fig. 8: Resource Utilization Report

VIII. Implementation and Timing Analysis

Post-route timing analysis was performed to determine the critical path delay and maximum operating frequency.

Design Timing Summary

Setup		Hold		Pulse Width	
Worst Negative Slack (WNS):	3.938 ns	Worst Hold Slack (WHS):	0.384 ns	Worst Pulse Width Slack (WPWS):	4.500 ns
Total Negative Slack (TNS):	0.000 ns	Total Hold Slack (THS):	0.000 ns	Total Pulse Width Negative Slack (TPWS):	0.000 ns
Number of Failing Endpoints:	0	Number of Failing Endpoints:	0	Number of Failing Endpoints:	0
Total Number of Endpoints:	16	Total Number of Endpoints:	16	Total Number of Endpoints:	33

All user specified timing constraints are met.

Fig. 9: Post-Route Timing Summary

MAXIMUM clock frequency: fmax= 1/Tmin = 1/(10-3.938ns) = 164.96 MHz

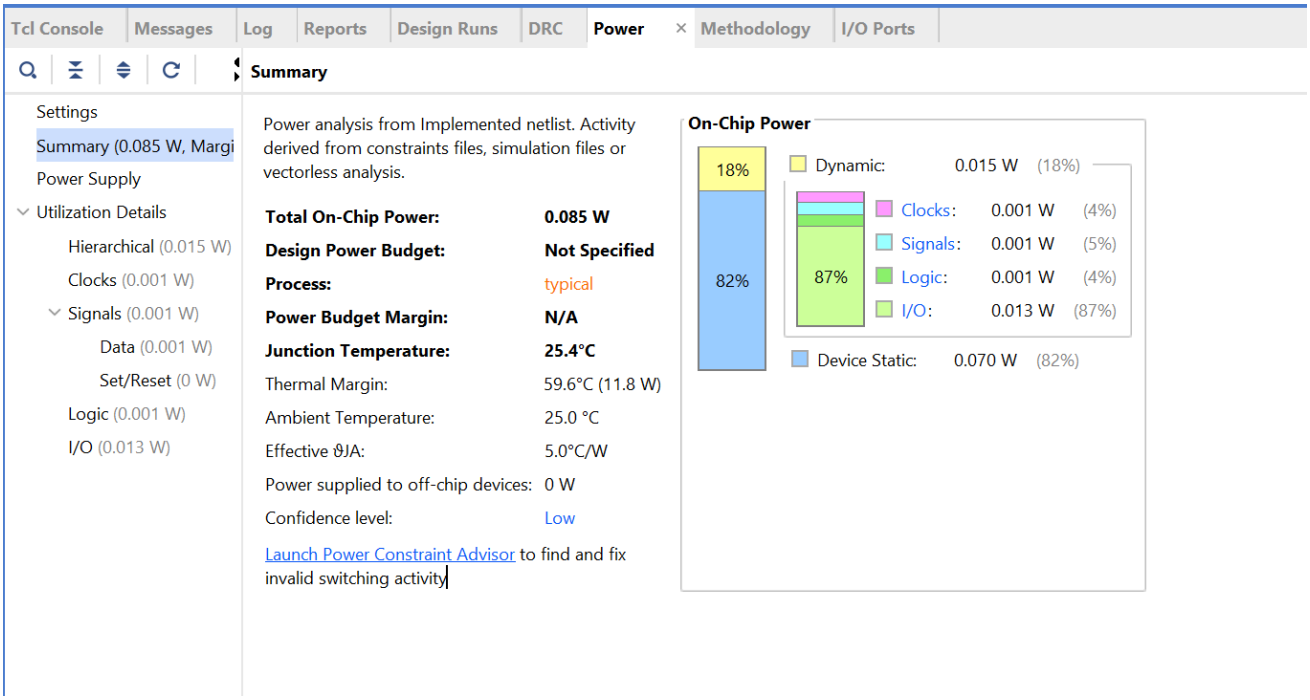


Fig. 10: Power Summary

IX. Discussion

Radix-4 Booth recoding reduces the number of partial products from eight to four. The use of Carry-Save Adders minimizes carry propagation delay by postponing final carry resolution. The critical path typically lies in the final carry-propagate stage or CSA reduction path depending on FPGA mapping.

X. Conclusion

A signed 8-bit Radix-4 Booth Recoded Multiplier was successfully designed and verified. Functional simulation confirmed correctness across extensive test cases. Synthesis and timing analysis provided insights into resource utilization and performance on Artix-7 FPGA.

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